# Project

## Synchronous FIFO

*Read the important notes and the requirements carefully at the end of the document.*

### Parameters

- FIFO_WIDTH: DATA in/out and memory word width (default: 16)
- FIFO_DEPTH: Memory depth (default: 8)

### Ports

| Port | Direction | Function |
|---|---|---|
| data_in | Input | Write Data: The input data bus used when writing the FIFO. |
| wr_en | | Write Enable: If the FIFO is not full, asserting this signal causes data (on data_in) to be written into the FIFO |
| rd_en | | Read Enable: If the FIFO is not empty, asserting this signal causes data (on data_out) to be read from the FIFO |
| clk | | Clock signal |
| rst_n | | Active low asynchronous reset |
| data_out | Output | Read Data: The sequential output data bus used when reading from the FIFO. |
| full | | Full Flag: When asserted, this combinational output signal indicates that the FIFO is full. Write requests are ignored when the FIFO is full, initiating a write when the FIFO is full is not destructive to the contents of the FIFO. |
| almostfull | | Almost Full: When asserted, this combinational output signal indicates that only one more write can be performed before the FIFO is full. |
| empty | | Empty Flag: When asserted, this combinational output signal indicates that the FIFO is empty. Read requests are ignored when the FIFO is empty, initiating a read while empty is not destructive to the FIFO. |
| almostempty | | Almost Empty: When asserted, this output combinational signal indicates that only one more read can be performed before the FIFO goes to empty. |
| overflow | | Overflow: This sequential output signal indicates that a write request (wr_en) was rejected because the FIFO is full. Overflowing the FIFO is not destructive to the contents of the FIFO. |
| underflow | | Underflow: This sequential output signal Indicates that the read request (rd_en) was rejected because the FIFO is empty. Under flowing the FIFO is not destructive to the FIFO. |
| wr_ack | | Write Acknowledge: This sequential output signal indicates that a write request (wr_en) has succeeded. |

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Overview of the testbench flow:

The top module will generate the clock, pass it to the interface, and the interface will be passed to the DUT, tb, and monitor modules. The tb will reset the DUT and then randomize the inputs. At the end of the test, the tb will assert a signal of size 1-bit named test_finished. The signal will be defined as well as the error_count and correct_count in a shared package that you will create named shared_pkg.

The monitor module will do the following:

1. Create objects of 3 different classes (FIFO_transaction, FIFO_scoreboard, FIFO_coverage). These classes will be discussed later in the document.
2. It will have an initial block and inside it a forever loop that waits for negedge clock at the start of the loop and then sample the data of the interface and assign it to the data variables of the object of class FIFO_transaction. Then after that there will be fork join, where 2 processes will run, the first one is calling a method named sample_data of the object of class FIFO_coverage and the second process is calling a method named check_data of the object of class FIFO_scoreboard.
3. So, in summary the monitor will sample the interface ports, and then pass these values through the FIFO_transaction object to be sampled for functional coverage and to be checked if the output ports are correct or not.
4. After the fork join ends, you will check for the signal test_finished if it is high or not. If it high, then stop the simulation and display a message with summary of correct and error counts.

**Steps:**

1. Adjust the design to take an interface and change the file extension to sv.
2. Create a package in a new file that will have a class named FIFO_transaction
    a. Inside of this class add the FIFO inputs and outputs as class variables of the class as well as adding 2 integers (RD_EN_ON_DIST & WR_EN_ON_DIST)
    b. Add a constructor that takes 2 inputs and override the values of RD_EN_ON_DIST and WR_EN_ON_DIST, let the default of RD_EN_ON_DIST be 30 and WR_EN_ON_DIST be 70
    c. Add the following 3 constraint blocks
        1. Assert reset less often
        2. Constraint the write enable to be high with distribution of the value WR_EN_ON_DIST and to be low with 100-WR_EN_ON_DIST
        3. Constraint the read enable the same as write enable but using RD_EN_ON_DIST
3. Create a package in a new file that will have a class for functional coverage collection named FIFO_coverage
    a. Import the previous package (Add the import statement after the package declaration)
    b. The class will have an object of the class FIFO_transaction named F_cvg_txn.
    c. Create a covergroup. The coverage needed is cross coverage between 3 signals which are write enable, read enable and each output control signals (outputs except data_out) to make sure that all combinations of write and read enable took place in all state of the FIFO so it is expected to have 7 total cross coverage inside of the covergroup.
    d. Create the covergroup inside the constructor.
    e. Create a void function inside it named sample_data that takes one input named F_txn. This input is an object of class FIFO_transaction. This function will do the following

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        1. Assign F_txn to F_cvg_txn
        2. Trigger the sampling of the covergroup using the .sample method

4. Create a package in a new file named FIFO_scoreboard
    a. Import the FIFO_transaction package.
    b. Add variables for the data_out_ref, full_ref, etc. to be used in the reference_model method
    c. Create a method named check_data that takes one input which of type FIFO_transaction
        1. Inside this method, call another method named reference_model that you will create and pass to it the same object that you have received
        2. Reference_model method will check the input values from the input object and assign values to the class properties data_out_ref, full_ref, etc.
        3. After the reference_model the method returns, you will compare the reference outputs calculated with the outputs of the object received. Increment the error_count or correct_count. Also, display a message if error occurs.
5. Open the design file and add assertions to the FIFO inside the design file.
    a. Add assertions to all the FIFO output flags (outputs except data_out) as well as the internal counters of the FIFO.
    b. **Extra part to be done:** Guard the assertions using conditional compilation with the `ifdef directive with macro named "SIM", and then include the macro in the vlog command +define+SIM option in the vlog command of your do file. Refer to this link to learn more about conditional compilation.

## Notes

1. Since assertions have been added to monitor the FIFO flags inside the design itself, the check_result method in your testbench will only need to verify the correctness of the data_out.
2. If both read and write enables are high simultaneously:
    a. When the FIFO is empty, only writing will take place.
    b. When the FIFO is full, only reading will take place.
3. On every reset assertion, all FIFO counters and pointers are reset. This means the FIFO is considered empty each time a reset occurs.
4. Use the event datatype to ensure the monitor samples the data before the testbench drives new values. Specifically:
    a. Trigger the event after you finish driving inputs and wait for the negedge clk.
    b. The monitor should wait for this event to be triggered and then sample the interface with both the inputs and the corresponding outputs.
5. Since class objects are created at simulation run time 0 in this testbench, they won't appear on the waveform unless you start the simulation. Here's a simple trick to add them on the wave from the beginning of the simulation:
    a. In the transcript, type "run 0". This ensures objects are initialized and visible in the "Objects" window.
    b. Drag them to the waveform window.
    c. Then type run -all to resume full simulation.
    d. You can automate this in your .do file to streamline the process.

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6. Here are some design requirements that the assertions need to check:
    a. Reset Behavior
        i. After reset is asserted (rst_n = 0), internal pointers and counters must reset to 0
    b. Write Acknowledge (wr_ack)
        i. When a write enable signal (wr_en) is active and the FIFO is not full, wr_ack should be asserted to confirm the write operation.
    c. Overflow Detection
        i. If a write is attempted when the FIFO is full, overflow should be asserted.
    d. Underflow Detection
        i. If a read is attempted when the FIFO is empty, underflow should be asserted.
    e. Empty Flag Assertion
        i. When the internal count is zero, the empty flag should be asserted.
    f. Full Flag Assertion
        i. When the internal count equals the FIFO depth, the full flag should be asserted.
    g. Almost Full Condition
        i. When the count reaches FIFO depth - 1, almostfull should be asserted.
    h. Almost Empty Condition
        i. When the count equals 1, the almostempty signal should be asserted.
    i. Pointer Wraparound
        i. After writing or reading FIFO_DEPTH entries (0 to 7), the write or read pointer should eventually wrap around back to 0.
    j. Pointer threshold
        i. Internal pointers cannot exceed the FIFO_DEPTH entries in any given time. Same applies for the counter.
7. Add assertions in the top file ensuring that asynchronous reset is working as expected.

## Requirements

1. Verification Plan, where you will list your verification plan, an example of the document can be found in the link here. In the functionality check column, you can specify whether the requirements are being checked only by the golden/reference model, assertion or both.
2. You are welcome to create your own constraints, coverpoints, and assertions. If you would like feedback or ideas, feel free to share your additions with me or your assigned teaching assistant.
3. Use Do file to run the top and make sure to generate the coverage report and check that the code coverage, functional coverage, and sequential domain (assertions) coverage are 100%
4. QuestaSim snippets, report any bugs detected and modify the RTL. Report the before and after to show the changes made to the RTL.

Submission file: **.rar file containing the following**:

- PDF file having the requirements with the bugs detected for the FIFO
- Do file to run the top
- SystemVerilog files & also add snippets to the code in the PDF, it will be easier for the grading process

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